

=== PHASE 1: CANONICAL TRUTH VALIDATION ===

Date: 2026-03-22

Scope: Active materials only (README.md, docs/core/*, docs/analysis/*, results/baseline/, results/analysis/)

Canonical sources: results/baseline/simulation/all_results_raw.csv , results/baseline/tables/*.csv

=== 1. INCONSISTENCIES FOUND ===

K-value Issues

No $K \neq 20$ usage issues in active baseline. $K=20$ is consistently used as the canonical baseline across all active docs. Fleet sensitivity analyses correctly use $K \in \{10, 15, 20, 25, 30, 35, 40, 45, 48\}$.

Capacity Issues

#	File	Issue	Severity
C1	docs/analysis/optimization_results.md	Marked  CURRENT but contains cap=5 optimization data. Table 2.1 shows Max Units=5 for P2 at $K \geq 30$, P2c with Max Units=5 at $K=20$. Header says “max 2 units” but all data is from cap=5 solver runs. The 46.5% improvement claim and all response time values (e.g., $P0=4.75$ at $K=20$) are optimization-only metrics from the cap=5 era — NOT simulation results.	CRITICAL
C2	docs/analysis/firehouse_capacity_analysis.md	Shows capacity=5 in code blocks. Correctly marked as  HISTORICAL with warning banner.	Minor (acceptable)

Numeric Inconsistencies

Technical Report (docs/core/technical_report.md) vs Canonical CSV

#	Metric	Tech Report Value	CSV Canonical Value	Location in Report	Severity
N1	P0 P90 RT	5.62 min	5.33 min	Lines 65, 421, 427, 714, 816, 900	CRITICAL
N2	P90 improvement (P2 vs P0)	-33.1%	-29.7%	Lines 65, 427, 714, 816, 900	CRITICAL (cascades from N1)
N3	P0 95% CI	[3.10, 3.24]	[3.14, 3.19]	Line 421	Stale
N4	P1 P90 RT	4.03	3.99	Line 422	Minor
N5	P0 Utilization	7.8%	7.6%	Lines 68, 421, 718	Stale
N6	P1 Utilization	7.5%	7.4%	Lines 68, 422	Minor
N7	P2 Utilization	7.5%	7.4%	Lines 68, 423	Minor
N8	P0 8-min Coverage	99.6%	99.7%	Lines 18, 67, 421, 428, 714, 734, 766, 816	Stale
N9	P2 8-min Coverage	99.6%	99.7%	Lines 18, 67, 423, 428, 714, 734, 766, 816	Stale
N10	P1 Mean RT	2.63	2.62	Lines 64, 422	Minor (rounding)
N11	P0 K=15 Mean RT	3.70	3.68	Line 464	Minor
N12	P0 K=30 Mean RT	2.78	2.81	Line 466	Minor
N13	P0 K=40 Mean RT	2.58	2.45	Line 467	Stale
N14	P0 K=15 Coverage	99.0%	99.1%	Line 464	Minor
N15		99.9%	99.8%	Line 466	Minor

#	Metric	Tech Report Value	CSV Canonical Value	Location in Report	Severity
	P0 K=30 Coverage				
N16	P0 K=40 Coverage	99.9%	99.8%	Line 467	Minor
N17	Utilization range	“7.5% to 7.8%”	7.4% to 7.6%	Line 718	Stale
N18	“ η^2 values above 0.99”	$\eta^2=0.959$ at K=20	CSV: F=1019, $\eta^2=0.959$	Line 714	Stale (over-stated)

README.md vs Canonical CSV

#	Metric	README Value	CSV Canonical Value	Severity
R1	P0 P95 RT	6.26	6.28	Minor
R2	P2 P95 RT	4.66	4.69	Minor
R3	P95 Improvement	−25.6%	−25.4%	Minor
R4	Mean RT Improvement	−19.0%	−18.9%	Minor
R5	P0 8-min Coverage	99.6%	99.7%	Minor
R6	P2 Utilization	7.5%	7.4%	Minor
R7	Unit test count	“39 unit tests (pytest)” (line 139), “39 tests” (line 303)	176 tests across 13 modules (pytest -co)	Stale
R8	Footer test count	“176 unit tests” (line 427)	176 tests ✓	OK
R9	Sim run count	“2,700+”	Tech report says 2,400; actual ~2,760 (see below)	Stale (all three documents disagree)

Executive Presentation (docs/core/executive_presentation.md) vs CSV

#	Metric	Exec Pres Value	CSV Canonical Value	Severity
E1	P0 P95 RT	6.26	6.28	Minor
E2	P2 P95 RT	4.66	4.69	Minor
E3	P1 P95 RT	5.05	5.01	Minor
E4	P0 6-min Coverage	93.7% (slide 2, line 30)	94.0%	Stale
E5	P0 8-min Coverage	99.6%	99.7%	Minor

Executive Summary (docs/core/executive_summary.md) vs CSV

- P90 values: 5.33, 3.99, 3.75 — **matches CSV** ✓
- 8-min Coverage: 99.7%, 99.6%, 99.7% — **matches CSV** ✓
- Mean RT: 3.17, 2.62, 2.57 — **matches CSV** ✓
- Sim run count: “2,700+” — disagrees with tech report “2,400”

Simulation Run Count Inconsistency

Document	Claimed Count
README.md	2,700+
Executive Summary	2,700+
Executive Presentation	2,700+
Technical Report (abstract, throughout)	2,400

Actual counts (from CSVs):

- all_results_raw.csv : 810 runs (3 policies × 9 K × 30 reps)
- exp1_policy_comparison.csv : 90 (overlap with above at K=20)
- exp2_fleet_sensitivity.csv : 720
- exp3_demand_sensitivity.csv : 540
- exp4_service_robustness.csv : 270
- cbd_experiment_results.csv : 330
- Production subtotal (exp1-4): 1,620
- Total non-duplicated: ~2,760 (baseline 810 + CBD 330 + remaining production unique ~1,620)

Neither “2,400” nor “2,700+” is precisely correct.

Artifact Issues

#	Issue	Severity
A1	<code>results/figures/capacity_sensitivity_heatmap.png</code> — Shows “Data format issue” placeholder text for K=20 and K=40 panels (broken figure, no actual heatmap data rendered). Per uploaded screenshot.	CRITICAL
A2	<code>results/analysis/figures/capacity_sensitivity_heatmap.png</code> — same issue as A1 (39KB, likely broken)	CRITICAL
A3	Tech report Figure 9 caption says “K=20, K=30, and K=40” but the actual heatmap only has K=20 and K=40 panels (K=30 panel missing)	Stale
A4	No <code>rt_heatmap_K30.png</code> exists in <code>results/analysis/capacity_comparison/</code> (only K20 and K40)	Stale

=== 2. CANONICAL VALUES ===

Source: `results/baseline/simulation/all_results_raw.csv` (K=20, cap=2, 30 reps)

Metric	P0	P1	P2
Mean Response Time	3.17 min	2.62 min	2.57 min
Mean RT 95% CI	[3.14, 3.19]	[2.60, 2.64]	[2.55, 2.59]
P90 Response Time	5.33 min	3.99 min	3.75 min
P95 Response Time	6.28 min	5.01 min	4.69 min
6-min Coverage	94.0%	98.0%	98.2%
8-min Coverage	99.7%	99.6%	99.7%
Mean Utilization	7.6%	7.4%	7.4%
Queue Fraction	0.0	0.0	0.0
Total Incidents (mean)	570.6	572.8	571.4

Derived:

- Mean RT improvement (P2 vs P0): **−18.9%**
- P90 RT improvement (P2 vs P0): **−29.7%**
- P95 RT improvement (P2 vs P0): **−25.4%**
- ANOVA F-statistic (K=20): **1,019.25** ($p < 0.001$, $\eta^2 = 0.959$)

Fleet sensitivity (P0, canonical):

K	Mean RT	8-min Coverage
15	3.68 min	99.1%
20	3.17 min	99.7%
30	2.81 min	99.8%
40	2.45 min	99.8%

Test count: 176 tests across 13 test modules

=== 3. AFFECTED DOCS/ARTIFACTS ===

File	Issues Found
docs/core/technical_report.md	N1-N18 (P90, CI, utilization, coverage, fleet sensitivity, η^2 claim)
README.md	R1-R9 (P95, improvement %, coverage, test count, run count)
docs/core/executive_presentation.md	E1-E5 (P95, coverage)
docs/core/executive_summary.md	Run count only (metrics are correct)
docs/analysis/optimization_results.md	C1 (entire document is cap=5 optimization data disguised as CURRENT)
results/figures/capacity_sensitivity_heatmap.png	A1 (broken figure)
results/analysis/figures/capacity_sensitivity_heatmap.png	A2 (broken figure)

=== 4. SEVERITY CLASSIFICATION ===

Critical (factual errors that change conclusions)

1. **N1+N2**: Tech report P0 P90 = 5.62 (actual: 5.33) → “33.1% P90 improvement” (actual: 29.7%). Appears in abstract, key findings table, results section, discussion, conclusions. Cascading error across 6+ locations.
2. **C1**: docs/analysis/optimization_results.md — marked CURRENT, contains cap=5 optimization-only data. All metrics (P0=4.75, 46% improvement, Max Units=5) are stale. This document is unreliable for anyone reading it as current.
3. **A1+A2**: capacity_sensitivity_heatmap.png — broken figure showing “Data format issue” placeholder. Referenced as Figure 9 in the tech report.

Stale (outdated numbers, small magnitude, don't change conclusions)

1. **N3**: P0 CI [3.10, 3.24] → [3.14, 3.19] (narrower range)
2. **N5+N17**: P0 utilization 7.8% → 7.6%; range “7.5%–7.8%” → “7.4%–7.6%”
3. **N8+N9**: 8-min coverage 99.6% → 99.7% (appears ~15 times in tech report)
4. **N13**: P0 K=40 RT 2.58 → 2.45
5. **N18**: “ η^2 values above 0.99” claim — actual $\eta^2=0.959$ at K=20 (still large, but overstated)
6. **R7**: “39 tests across 4 modules” → 176 tests across 13 modules
7. **R9**: Simulation run count: README/exec say 2,700+, tech report says 2,400 (actual ~2,760)
8. **E4**: P0 6-min coverage 93.7% → 94.0% (one location in exec pres)
9. **A3**: Tech report Figure 9 caption says K=20,K=30,K=40 but figure only has K=20,K=40

Minor (rounding differences ≤ 0.05 , don't affect narrative)

1. **N4,N6,N7,N10,N11,N12,N14,N15,N16**: Small rounding differences (≤ 0.04 min RT, ≤ 0.1 pp coverage)
 2. **R1-R6**: README rounding differences (P95 off by 0.02-0.03, coverage off by 0.1pp)
 3. **E1-E3,E5**: Exec presentation rounding differences
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END OF PHASE 1 VALIDATION REPORT — NO FIXES APPLIED